

Market Microstructure and Optimal Execution in Thin Crypto Markets: Evidence from iExec RLC

QuantFi

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Abstract

We utilize high-resolution limit-order-book data—comprising top-of-book snapshots, Level-2 deltas, and complete trade tapes across seven venues—to characterize the market microstructure of a thinly traded cryptocurrency (iExec RLC) and derive the optimal liquidation schedule for a large block position. Empirical analysis reveals a structurally thin consolidated order book where displayed depth is substantially inflated by transient quoting; near-touch quote churn reaches up to 15,000 times traded volume, necessitating a 14% haircut to establish executable firm depth. Market making is heavily concentrated and venues are tightly coupled. Price discovery is led by a single primary exchange, with cross-venue contagion severely depleting liquidity for aggressive trades exceeding 1,000 RLC. By modeling dealer inventory as a mean-reverting process, we estimate a sustainable market absorption rate of 484 RLC per minute and identify robust cross-venue inventory dispersion.

Applying these microstructure dynamics to the liquidation of a \$1,000,000 position, representing 3.5 times the asset’s daily trading volume, we demonstrate that the instantaneous firm book cannot absorb even a third of the target. Without access to over-the-counter liquidity, the optimal strategy mandates algorithmic dispersion across exchanges in small clips paced strictly below the dealer absorption rate, forcing a minimum execution horizon of 13 to 14 days. Ultimately, we bound the total liquidation cost between 591 and 2,695 basis points. We conclude that execution frictions account for a fractional 230 basis points, while the dominant, un-hedgeable risk remains exogenous price drift during the mandatory holding period.

1 Theoretical Framework

This section establishes the theoretical foundation for our empirical analysis by reviewing core market microstructure frameworks [4, 12]. We formalize the requisite mathematical notation here to ensure all subsequent results remain self-contained.

1.1 Liquidity, displayed depth, and firm depth

A limit order book at instant t for one market is the set of resting bids $\{(p_i, q_i)\}$ (price, size), ordered $p_1 > p_2 > \dots$. The *cost to sell* a quantity Q by walking the bids is the volume-weighted shortfall against a reference (“fair”) price p_0 :

$$s(Q) = \frac{p_0 - \bar{p}(Q)}{p_0}, \quad \bar{p}(Q) = \frac{\sum_i p_i \min(q_i, r_i)}{Q}, \quad (1)$$

where r_i is the residual size filled at level i and $\bar{p}(Q)$ is the average fill price. We report $s(Q)$ in basis points (1 bps = 0.01%) and take p_0 to be the robust consolidated touch (the median of venues’ best bids), which is insensitive to a single stale or mis-converted quote.

Displayed depth need not be *firm*. Many resting quotes are cancelled within milliseconds (“flicker”), so the depth an arriving order actually meets is smaller than the snapshot shows. We define the firm fraction of near-touch depth as the time-weighted share of displayed size that has rested at least τ_f seconds,

$$\phi = \frac{\text{depth resting} \geq \tau_f}{\text{depth displayed}}, \quad \tau_f = 1 \text{ s}, \quad (2)$$

and the difference $1 - \phi$ is the “ghost” haircut. A complementary statistic is the *churn ratio*, the quantity of near-touch liquidity posted-and-pulled per unit actually traded; a churn of C means C units are quoted for every one consumed.

1.2 Market making, adverse selection, and inventory

Dealers quote a two-sided spread to earn the bid–ask while bearing two risks. The first is *adverse selection*: informed counterparties pick off stale quotes, so, in the tradition of Kyle [10] and Glosten and Milgrom [7], the spread widens with the probability and magnitude of informed flow—empirically, with volatility. We summarise this by a linear policy

$$\text{spread}_t = a + b \sigma_t, \quad (3)$$

with σ_t a short-horizon realized volatility and $b > 0$ the adverse-selection slope.

The second is *inventory risk*. Following Ho and Stoll [9] and Avellaneda and Stoikov [3], a dealer holding inventory X skews quotes to mean-revert toward a target: as X grows long, the dealer lowers (skews down) its bid to discourage further buying and encourage selling. We model the aggregate inventory the market carries as an Ornstein–Uhlenbeck (OU) process,

$$dX_t = \theta (\bar{X} - X_t) dt + \sigma_X dW_t, \quad (4)$$

where $\bar{X}$ is the target, θ the mean-reversion speed, and W_t a Wiener process. The half-life of a disturbance is $t_{1/2} = \ln 2 / \theta$, and the *sustainable absorption rate*—how fast the market can take inventory before quotes thin—is the swing amplitude divided by the rebalancing time,

$$\text{absorption} \approx \frac{\text{inventory swing}}{t_{1/2}} \quad [\text{RLC/min}]. \quad (5)$$

A “flip point” is the inventory level at which expected dealer flow changes sign (from absorbing to off-loading); identifying it requires observing inventory actually cycle, which a short, single-regime sample may not deliver.

1.3 Cross-venue price discovery and contagion

When an asset trades on many venues, prices are kept in line by arbitrage, but not instantaneously. One venue typically *leads*: its mid moves first and the others follow with a lag [8]. A market sell that consumes bids on the lead venue therefore (i) moves that venue’s mid and (ii) propagates to the others, which re-quote lower and may pull their own bids—a contagion that makes the consolidated book less additive than it appears. We measure propagation by an event study around sell prints, comparing the post-print move on *other* venues to a pre-print placebo window (the churn baseline), so that only the print-induced excess is attributed to contagion.

1.4 Optimal execution with drift, and the two-drift decomposition

A large sell cannot be done at once without ruinous impact, nor spread out indefinitely without bearing price risk. Almgren and Chriss [2] formalise the trade-off (see also Almgren [1] for nonlinear impact and Obizhaeva and Wang [11] for order-book supply/demand dynamics): liquidating Q over horizon T in increments x_t incurs temporary impact (a cost on each clip) and timing risk (variance of the unsold position), and the optimal schedule minimises a mean–variance objective

$$\min_{\{x_t\}} \underbrace{\sum_t \eta g(x_t)}_{\text{impact}} + \underbrace{\lambda \sigma \sqrt{\sum_t (\sum_{u>t} x_u)^2}}_{\text{timing risk}}, \quad (6)$$

with η an impact coefficient, $g(\cdot)$ the impact function, σ volatility, and λ risk aversion. Classical Almgren–Chriss assumes zero drift. A declining asset adds a third term: every unit of inventory held while the price falls is a loss. We therefore separate *two drifts*:

$$\text{drift cost} = \underbrace{\text{permanent impact}}_{\text{endogenous: we push the price down}} + \underbrace{\text{exogenous trend } \mu}_{\text{the asset falls regardless}}. \quad (7)$$

For a program of horizon T sold at roughly uniform rate against daily volume V , a transparent reduced form for the all-in cost (in bps of notional) collects an impact term that scales with participation $\rho = Q/(VT)$, the timing-risk term, and the exogenous-drift term:

$$C(T) = \underbrace{(1 + \tfrac{1}{2}f) Y \sigma \sqrt{\tfrac{Q}{VT}}}_{\text{temporary+permanent impact}} + \underbrace{\left(1 - \frac{e^{\mu T} - 1}{\mu T}\right)}_{\text{exogenous drift}} + \underbrace{\lambda \sigma \sqrt{T/3}}_{\text{timing risk}}, \quad (8)$$

where f is the permanent fraction of impact (Eq. 7), Y an order-of-unity impact constant, and all terms are expressed in bps. The impact term embodies the *square-root law* of market impact [6, 13]: the cost of trading a quantity scales as $\sqrt{Q/V}$,

$$\text{impact} \approx Y \sigma \sqrt{Q/V}, \quad (9)$$

which we use as a model-independent cross-check on Eq. 8 because it requires only volatility and volume, not order-book reconstruction.

2 Data

Venues, instruments, and streams. We record iExec RLC across seven centralised venues. For each market the collector captures three streams over WebSocket: (i) top-of-book *snapshots* on a fixed timer; (ii) the full *Level-2 delta stream*, every change to a price level; and (iii) the complete *trade tape*. Reconstructing the book at any instant—nearest snapshot as anchor, deltas replayed—reproduces the independently recorded snapshots to 98.5–100% at the touch and 99.8–100% at the level, which validates the reconstruction layer that all delta-based statistics rest on.

Window and what it can support. The analysis window is 2026-06-03 to 2026-06-07 UTC, ≈ 4.5 days of order book (the trade tape spans 5.4 days), comprising ≈ 1.89 million snapshots, ≈ 19 million unique L2 deltas, and 36,689 trades (binance’s tape backfilled to completeness from the venue’s public archive). Table 1 summarises the sample. This is enough to characterise microstructure and quoting behaviour in the prevailing regime, and to bound execution cost; it is *not* enough to forecast the price path, to identify behaviour in a stress regime, or to pin the absolute level of dealer inventory. We flag such limits throughout and consolidate them in Section 6.

Two data-integrity corrections. Two corrections matter for every consolidated figure. First, coinbase’s RLC/USD and RLC/USDC feeds are byte-identical (the same 1,634 trades, 100% overlap)—a collector artifact—so coinbase is counted *once*; this de-duplicates depth, volume, and inventory and makes every consolidated number conservative. Second, bitget’s delta stream re-anchors on checksum errors and is excluded from all delta-based statistics. The kucoin/BTC leg is excluded from the consolidated USD ladder because its implied BTC/USD conversion is a window-level scalar and is unreliable intraday.

A consistency caveat on method. Two code paths touch the same data: a *snapshot-direct* path on a 10-second grid (used for the consolidated depth and the model ladder) and a *full delta-reconstruction* path (used for the per-venue quote-dynamics and MM tools). They differ slightly by construction. Where a quantity is method-dependent we state the method and do not splice sources; the most consequential case is aggregate MM inventory (Section 4.4).

Table 1: Data sample. RLC spot markets, 2026-06-03 to 2026-06-07 UTC.

Item	Value
Venues (consolidated USD ladder)	binance, coinbase, gate, kraken, kucoin, cryptocom
Markets (after de-duplication)	9 spot markets; bitget excluded from delta steps
Order-book snapshots	$\approx 1,892,780$ (10 s grid)
Level-2 delta events	≈ 19 million
Trades	36,689
Window length	≈ 4.5 days (order book); 5.4 days (trade tape)
Reconstruction accuracy	98.5–100% (touch), 99.8–100% (level)
Reference mid p_0	\$0.3459

3 Methodology

Each microstructure quantity is estimated by a dedicated, self-verifying tool: depth and cost-to-trade by walking the FX-normalised consolidated ladder; firmness, quote lifetime, flicker and churn from the delta stream; the MM census, spread/skew policy and inventory from quote and signed-trade dynamics; absorption and the flip point by fitting Eq. 4; lead-lag and contagion by the cross-venue event study; and impact recovery from post-print mid trajectories. Every tool ships unit and synthetic-recovery tests—for example, the OU estimators recover a known target, half-life and flip point from simulated paths and report “no mean-reversion” on a random-walk control—and the full stack passes 28/28 end-to-end checks.

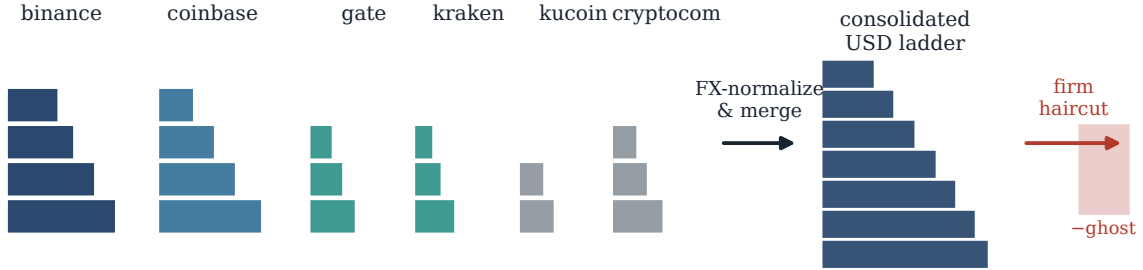
Consolidation. Per-venue books are converted to USD using FX implied from the data (EUR $\approx$ 1.160, BTC $\approx$ \$61.7k) and merged into one ladder, then haircut to firm depth via Eq. 2. Figure 1 is the schematic.

The model ladder. To lift microstructure into an execution cost we relax one assumption at a time, measuring the marginal effect on a single tracked quantity (Figure 2). *Phase I* (static) asks how much can be raised at a single instant: M0 frictionless $\rightarrow$ M1 single venue $\rightarrow$ M2 consolidated $\rightarrow$ M3 firm $\rightarrow$ M4 net of fees. The tracked quantity (net USDC at one instant) collapses from \$1.00M to $\approx$ \$0.28M, which *derives*—rather than asserts—that the problem is time-distributed.

Phase II (dynamic) then adds time and participation (M5), the reactive counterparty (M6), the temporary/permanent impact split (M7), the speed/risk frontier with drift (M8), cross-venue coupling (M9), and synthesis with the square-root cross-check (M10).

Bounding contagion scaling. The contagion event study is calibrated on the prints that exist in the sample, which are small (~ 300 RLC median). Its scaling to the larger clips of an execution program is *not identified*; we therefore carry the per-clip cross-venue term as a swept band (a clip-size exponent from 0, contagion does not grow, to $\frac{1}{2}$, square-root growth) rather than a point estimate, and we cross-check the impact term with the order-book-free square-root law (Eq. 9).

Schematic: from fragmented venue books to a firm consolidated ladder



Per-venue books (heights $\propto$ firm depth) are FX-normalized to USD, merged into one consolidated ladder, then haircut to firm (rested) depth — the basis of the M2 $\rightarrow$ M3 cost curve.

Figure 1: From fragmented venue books to a firm consolidated ladder. Per-venue books (heights $\propto$ firm depth) are FX-normalised to USD, merged into one ladder, then haircut to firm (rested) depth—the basis of the M2 $\rightarrow$ M3 cost curve in Figure 3. Schematic.

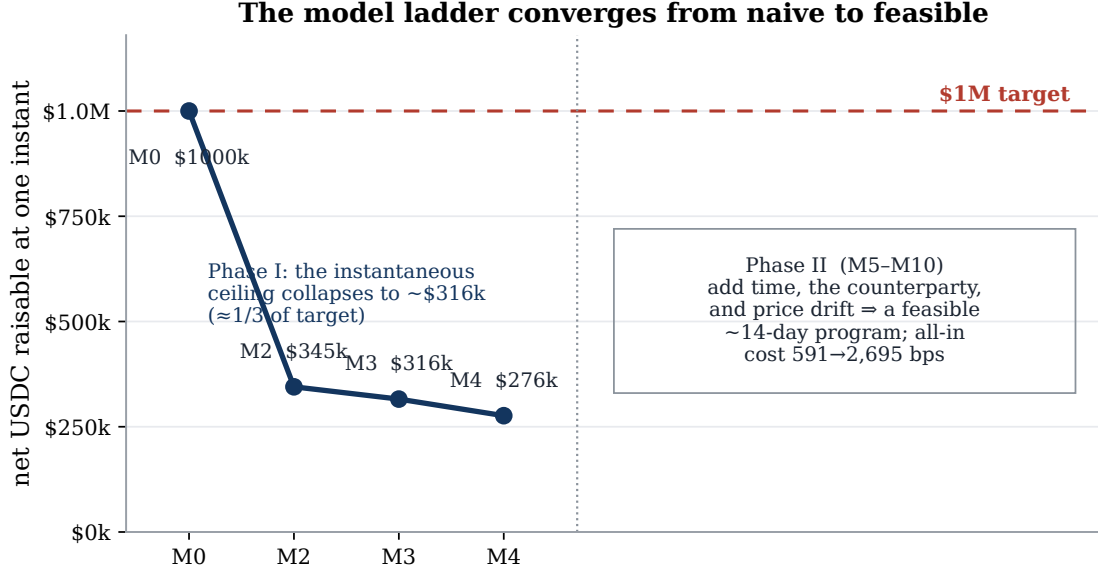


Figure 2: The model ladder. The tracked quantity (net USDC raisable at a single instant) collapses to roughly one third of target across Phase I (M0–M4); Phase II then prices the unavoidable multi-day program.

4 Empirical Study: Microstructure and Market-Maker Behaviour

This section is the core contribution. We characterise, in turn, consolidated depth and cost-to-trade; firm versus ghost liquidity and quote dynamics; the market makers—their census, policy, and inventory; cross-venue coupling; and intraday structure. The liquidation in Section 5 then follows from these facts.

4.1 Consolidated depth and the cost to trade

Figure 3 and Table 2 report the volume-weighted slippage (Eq. 1) as a function of order size across three environments: a single venue (M1), the consolidated visible book (M2), and the consolidated firm book (M3). Three structural characteristics emerge. First, even the deepest individual venue (Binance) lacks substantial capacity; executing a 250,000 RLC order incurs 808 bps of slippage and fully depletes the available firm depth. Second, because liquidity is fragmented, consolidating the order book significantly reduces execution costs for smaller trades. For instance, a 100,000 RLC order costs 38 bps on the visible consolidated book, compared to 172 bps on Binance alone. Third, despite the benefits of consolidation, the aggregate book remains structurally thin and sharply convex. Execution costs are relatively benign up to 500,000 RLC (166 bps on the firm book) but escalate rapidly, reaching 2,007 bps for a 1,000,000 RLC order. An order of 2.78 million RLC entirely exhausts the available depth across all venues. Consistent with Cont et al. [5], this convexity arises because liquidity is highly concentrated at the touch; once these primary levels are consumed, subsequent units execute against deep, sparsely populated price levels.

To quantify the market’s instantaneous capacity, we sample the firm consolidated book across 300 distinct intervals over the observation window (Figure 4). The median absorbable notional value is \$315,700, with a 10th to 90th percentile range of \$289,700 to \$356,300. Even at peak observed liquidity (\$380,800), the consolidated book can clear less than 40% of the \$1,000,000 liquidation target. This severe capacity constraint forms the empirical foundation for our time-distributed execution model.

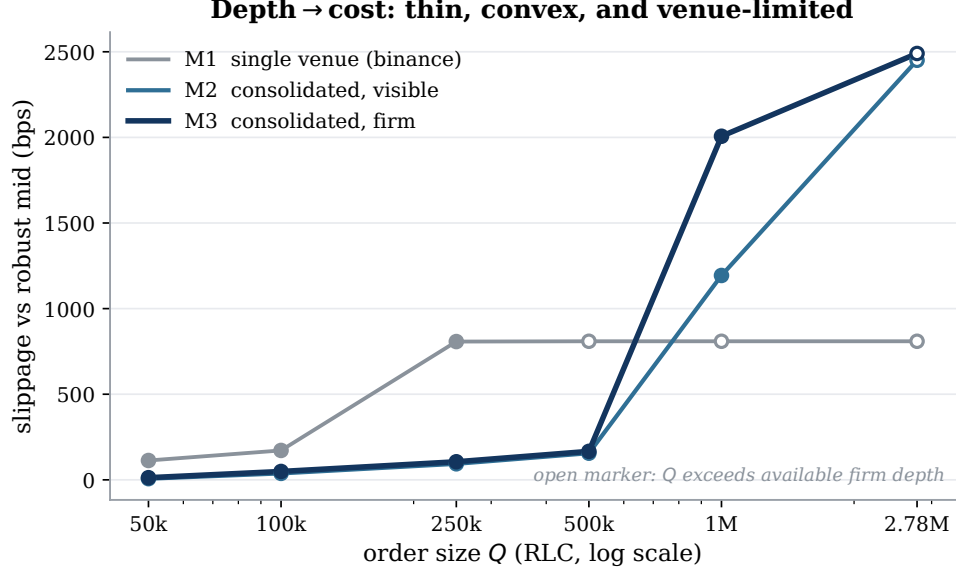


Figure 3: Depth→cost. Slippage (Eq. 1) versus order size for single-venue, consolidated-visible, and consolidated-firm books. Open markers: the order exceeds available firm depth. The curve is thin and convex; cost is benign to $\sim 500k$ RLC and explodes by 1M.

Table 2: Depth–cost curve: slippage (bps) versus order size, by liquidity definition. * order exceeds available firm depth.

Order (RLC)	50k	100k	250k	500k	1M	2.78M
M1 single venue (binance)	113	172	808	809*	809*	809*
M2 consolidated, visible	7	38	94	156	1,194	2,450*
M3 consolidated, firm	13	49	106	166	2,007	2,490*

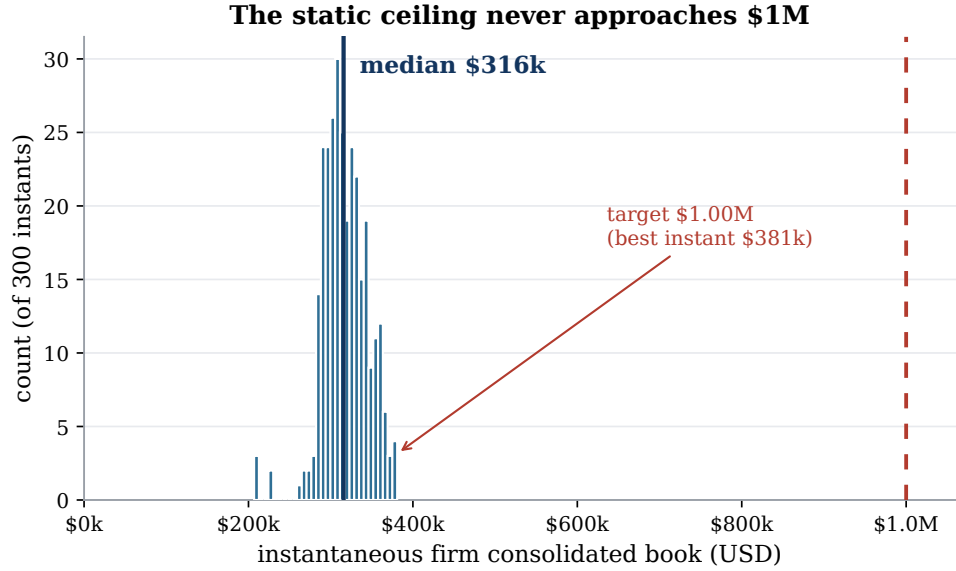


Figure 4: Distribution of the instantaneous firm consolidated book over 300 sampled instants. Median \$315.7k; the \$1M target is never reached.

4.2 Firm versus ghost liquidity, and quote dynamics

Displayed depth flatters the book. Table 3 and Figure 5 report, per venue, the firm fraction (Eq. 2), the median quote lifetime, and the churn ratio. The picture is sharply heterogeneous. binance is the anchor: 98% of near-touch depth is firm, quotes rest a median of 6.7 s, and only 56 units are quoted per unit traded. At the other extreme, coinbase displays the most size (the deepest near-touch book) but only 76% is firm, quotes live under one second, and churn is $\sim 4,000\times$; cryptocom and the kucoin pairs churn $6,600\times$ and $15,000\times$. Consolidated, the firm haircut removes $\approx 14\%$ of displayed near-touch depth—but the haircut is concentrated on exactly the venues that look deepest, so it reshapes *where* executable liquidity sits, not merely how much there is. Operationally, a marketable order that “sees” coinbase’s large displayed bids will, on arrival, often find a thinner, flickering book; binance’s smaller displayed size is more dependable.

Table 3: Quote dynamics by venue (near touch). Firm fraction (Eq. 2), median quote lifetime, churn ratio (quoted per traded), and the dealer census: effective distinct quote sizes, coordinated-ladder fraction, and refresh cadence.

Venue	Firm %	Quote life	Churn	Eff. sizes	Coord. %	Refresh
binance	98	6.7 s	$56\times$	62.5	70	300 ms
gate	94	0.53 s	$723\times$	31.3	86	93 ms
kraken/USD	97	2.1 s	$236\times$	37.1	45	21 ms
kraken/EUR	81	1.8 s	$1,259\times$	22.0	41	21 ms
kucoin/USDT	93	3.9 s	$2,275\times$	41.8	5	99 ms
kucoin/BTC	94	3.2 s	$15,365\times$	51.8	3	244 ms
coinbase	76	0.9 s	$4,040\times$	37.0	90	95 ms
cryptocom	81	13.7 s	$6,654\times$	28.0	91	1,009 ms

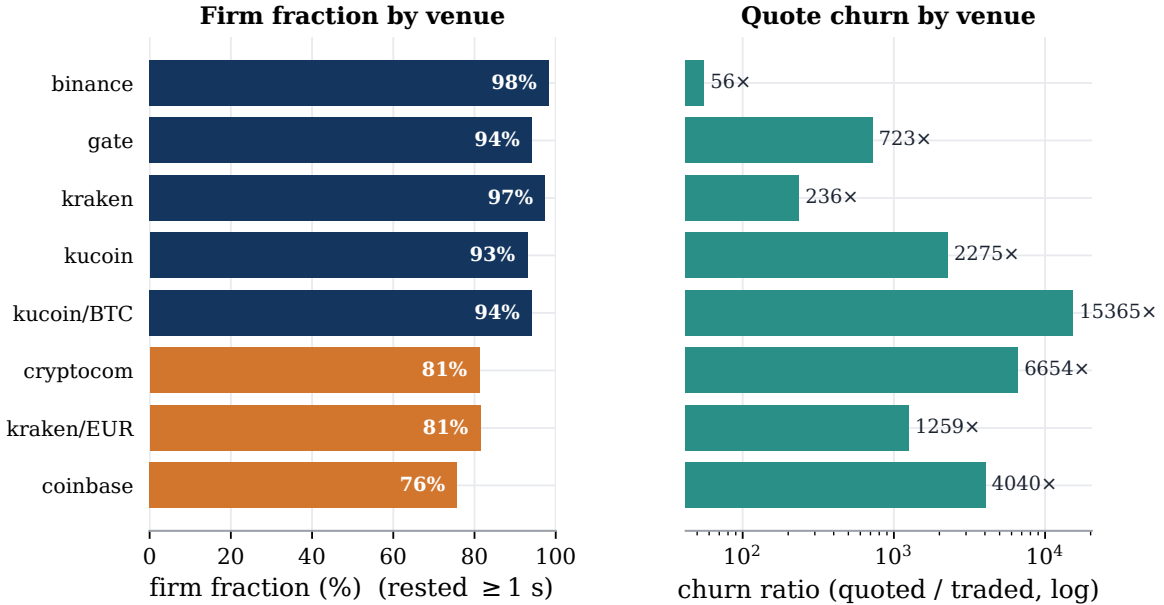


Figure 5: Firm fraction (left) and quote churn (right, log scale) by venue. binance is firm with low churn; coinbase, cryptocom and kucoin display size but churn heavily.

4.3 The market makers: census and policy

Public L2 data carry no order identifiers, so we cannot count dealers; we can, however, bound concentration and read the representative policy (Table 3, right columns). The effective number of distinct quote sizes is small (~ 22 – 62 per venue) and the coordinated-ladder fraction—the share of updates that move many price levels in one event, the signature of a single algorithm re-quoting an entire ladder—is high on most venues (70% on binance, 86–91% on gate/coinbase/cryptocom), with kucoin the competitive exception ($\leq 5\%$). This is a concentration lower bound: a few algorithms supply most quotes [not identified] as a head count, [directional] as a concentration picture.

The representative policy is adverse-selection-aware (Figure 6). Estimating Eq. 3 per venue, the spread–volatility slope b is positive everywhere it is defined—0.52 bps per $\text{bps}/\sqrt{\text{min}}$ on binance, 1.54 on coinbase, 3.09 on kraken/USD—with a cross-venue median of ≈ 1.10 . Dealers widen when volatility (and thus the risk of being picked off) rises, exactly as Eq. 3 predicts. The refresh cadence spans three orders of magnitude, from 21 ms on kraken to ~ 1 s on cryptocom, indicating a mix of latency-sensitive and slower quoting engines.

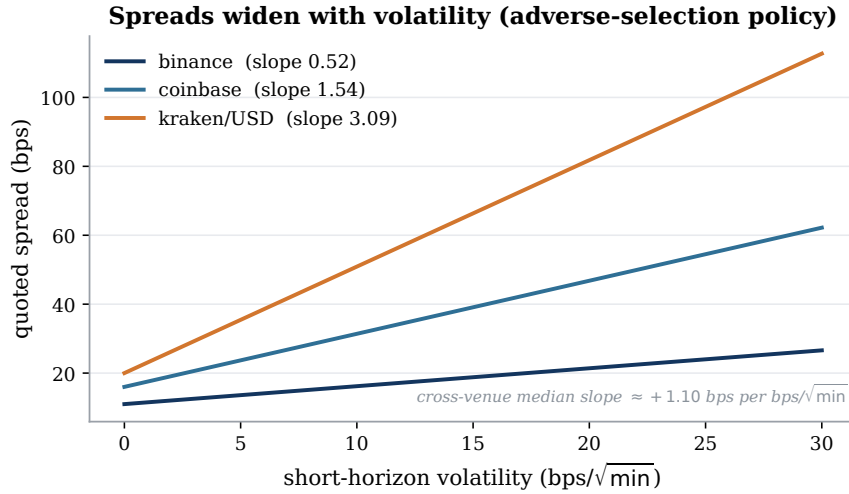


Figure 6: Adverse-selection policy: quoted spread widens with short-horizon volatility (Eq. 3), illustrated with the estimated per-venue slopes.

4.4 Inventory, absorption, and the (unidentified) flip point

Dealer inventory governs how much selling the market will absorb before bids thin. Two findings are robust and one is not.

Robust: the absorption rate. Fitting the OU process (Eq. 4) to the consolidated inventory gives a rebalancing half-life of ≈ 7.6 min and a short-run inventory swing of ± 3.7 k RLC, hence a sustainable absorption of ≈ 484 RLC/min (Eq. 5; bootstrap range across sub-windows 228–775). The delta-reconstruction path gives a consistent ≈ 588 RLC/min. Figure 7 illustrates the mechanism: inventory mean-reverts, and the rate at which it can do so caps how fast the market clears flow. [identified]

Robust: the cross-venue dispersion of inventory. Dealers are not uniform. The de-duplicated venue-split (Figure 8, Table 4) shows binance MMs net *short* (-202 k RLC) and kraken/USD short (-87 k), while coinbase MMs are saturated *long* ($+154$ k) and kraken/EUR long ($+98$ k). A short dealer is a natural buyer (it wants inventory back to target) and a saturated-long dealer is a reluctant

one—directly relevant to where selling is cheapest. [directional]

Not identified: the level and the flip point. The *level* of aggregate inventory depends on the origin of the cumulative signed-flow integration and on the de-duplication choice: the snapshot path gives +80.6k RLC, the de-duplicated venue-split sums to −29k, and a non-de-duplicated path gives +284k. Because the level is not pinned, neither is the “flip point” (the level at which dealers turn from buyers to sellers): the estimator recovers a known flip from synthetic data, but the real flip is only identified once inventory is observed to cycle through it, which this calm, single-regime sample does not deliver. We therefore use the absorption *rate* and the inventory *dispersion*, and we decline to lean on any inventory level or flip threshold. [not identified]

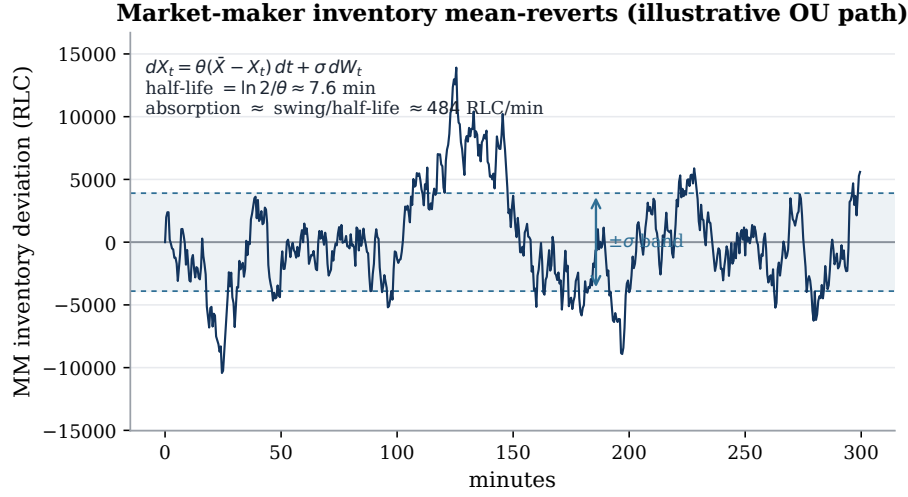


Figure 7: Market-maker inventory mean-reverts (illustrative OU path with the estimated half-life and swing). Absorption $\approx$ swing/half-life ≈ 484 RLC/min (Eq. 5).

Inventory is venue-split: binance MMs short, coinbase MMs saturated long

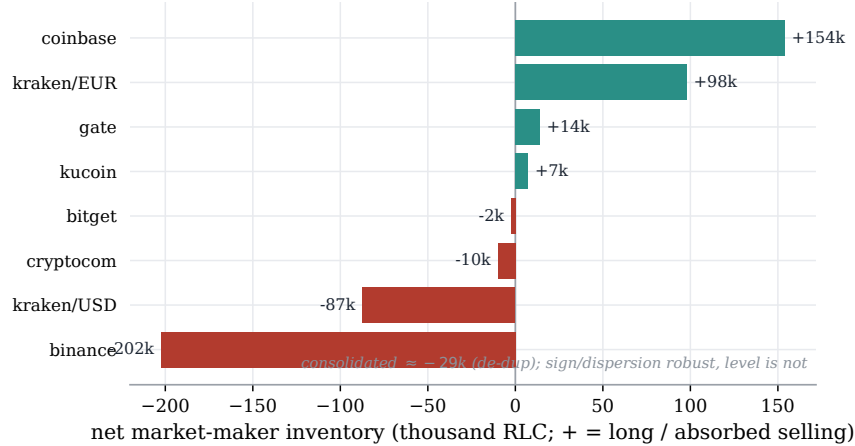


Figure 8: De-duplicated net MM inventory by venue. binance dealers run short (natural buyers); coinbase dealers are saturated long. The sign/dispersion is robust; the consolidated *level* ($\approx -29k$ here) is method-dependent and not identified.

Table 4: Net MM inventory by venue (de-duplicated; + = long/absorbed selling). Consolidated ≈ -29 k RLC under this de-duplication; +80.6k under the snapshot signed-flow path—the level is method-dependent. Source: `analysis_out/mmdeep_final.txt`; `mm.json`.

Venue	Net inventory (RLC)
binance	-202,328
kraken/USD	-87,407
cryptocom	-9,951
bitget	-2,385
kucoin (USDT+BTC)	+7,251
gate	+13,897
kraken/EUR	+98,036
coinbase	+153,942
Consolidated (de-dup)	-28,945

4.5 Cross-venue coupling: lead-lag, contagion, recovery

Venues are tightly linked (Figure 9). On a single clock built from the delta stream, binance leads price discovery and the others follow with a 250–500 ms lag (gate is a co-leader). The execution event study confirms causation: when a venue is hit, its own mid moves +8.3/ +8.7 bps at +1/ +5 s while the *other* venues move +5.3/ +6.4 bps, far beyond their pre-print placebo of ≈ 1 bp. At small print sizes the others follow in *price* but do not visibly pull depth.

Depth-pull is size-dependent, and we measure it net of churn (Figure 10). The probability that the worst-hit other venue’s bid falls by half is already ~ 30 –45% before any print—pure churn—so the contagion signal is the *excess* over that baseline. The knock-on mid move on other venues roughly doubles as clips cross $\sim 1,000$ RLC (from ~ 3.5 bps net at small sizes to ~ 7 bps at 1–2k), and the net depth-pull turns clearly positive in the same range; for the largest prints in the sample (4–10k RLC) individual episodes drive a single other venue’s bid to near zero within seconds. The practical implication is a clip-size threshold: keep clips below $\sim 1,000$ RLC to stay in the regime where cross-venue impact is small and churn-dominated rather than print-driven.

Finally, the cross-venue book *rebuilt*s over tens of seconds (Figure 11): after a sell, the average other-venue depth returns to its no-shock level within ~ 10 –30 s, while the single worst-hit venue takes longer. This sets the minimum safe *cadence* between large clips—wait long enough for the consolidated book to rebuild rather than compounding the pull.

4.6 Permanent versus temporary impact, and intraday structure

Not all impact recovers. Tracking the mid after sell prints on binance (Figure 12; 2,424 prints, median 269 RLC), displacement peaks at ≈ 10.9 bps and decays only to ≈ 7.3 bps by +300 s—a *permanent fraction* of $f \approx 0.67$ (Eq. 7). This is partly entangled with the prevailing downtrend (the pre-print placebo is -2.5 bps), so f is best read as “most impact is durable” [directional] rather than a precise constant; large-clip permanence is unidentified for want of large prints. Intraday, liquidity has a clear regime (Figure 13): cost-to-trade is lowest and firm depth deepest around 07:00 UTC, with a secondary deep window midday–evening, motivating timing the program into deeper hours.

Mechanism: a hit on the price leader propagates within seconds

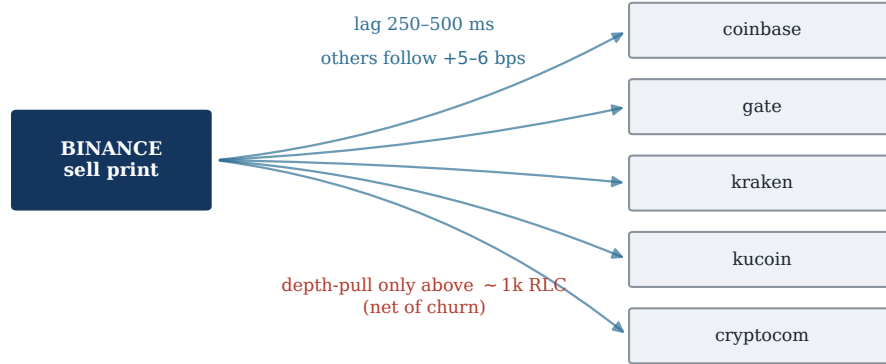


Figure 9: Cross-venue mechanism: a sell on the price leader (binance) propagates to the other venues within 250–500 ms, which follow +5–6 bps; depth-pull becomes material only above ~1k RLC.

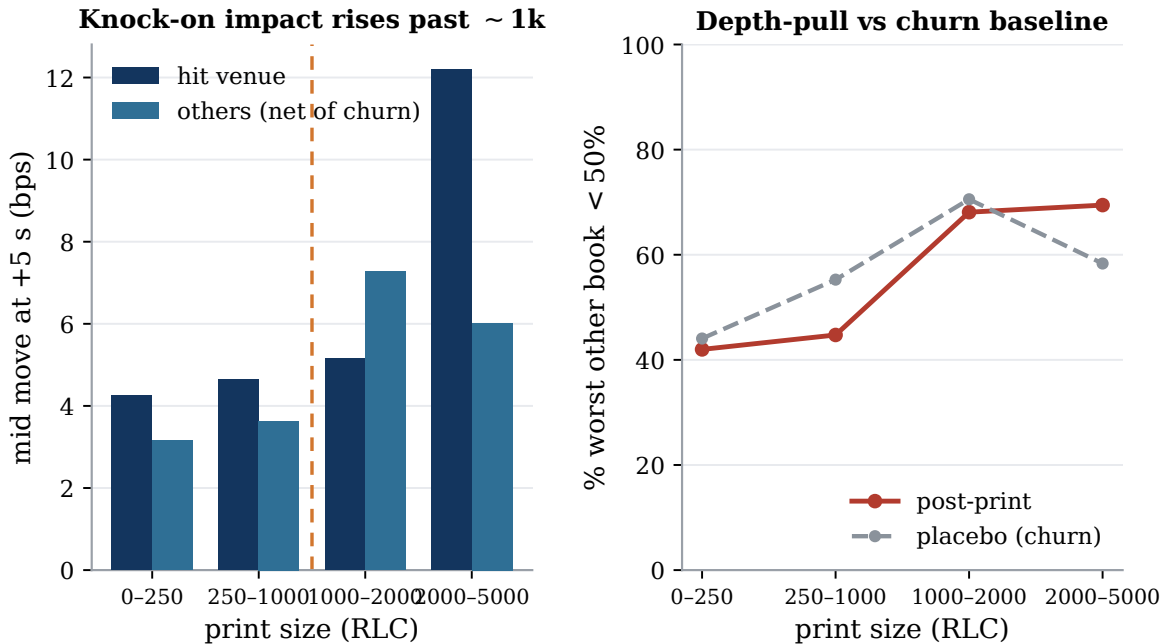


Figure 10: Contagion by clip size, net of churn. Left: knock-on mid move on other venues roughly doubles past ~1k RLC. Right: the “books vanish” depth-pull is largely churn at small sizes; the net excess grows with size.

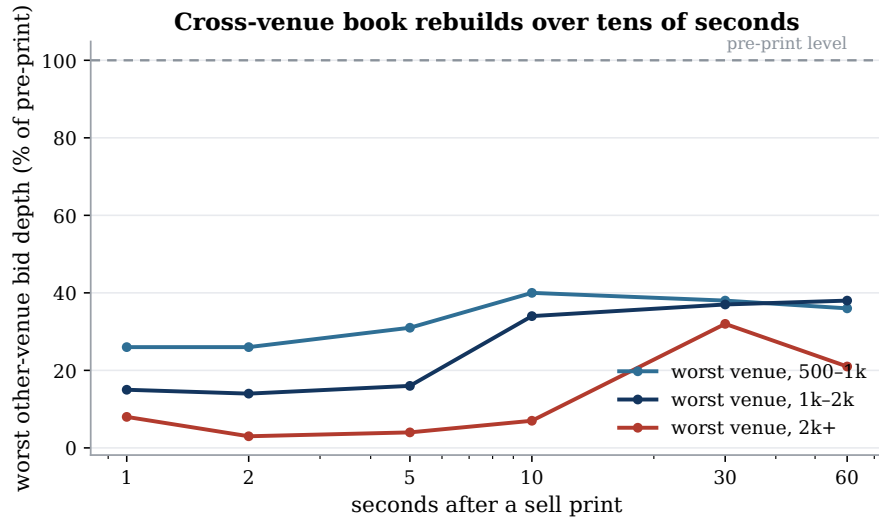


Figure 11: Cross-venue bid recovery after a sell print: the worst-hit other venue’s depth versus horizon, by size. Recovery over tens of seconds sets the minimum clip cadence.

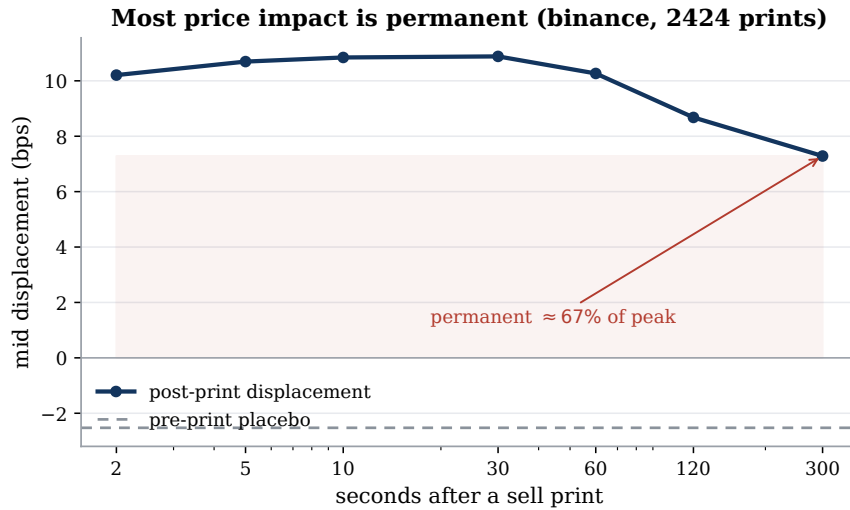


Figure 12: Impact is mostly permanent: post-print mid displacement decays only modestly over five minutes, a permanent fraction $f \approx 0.67$.

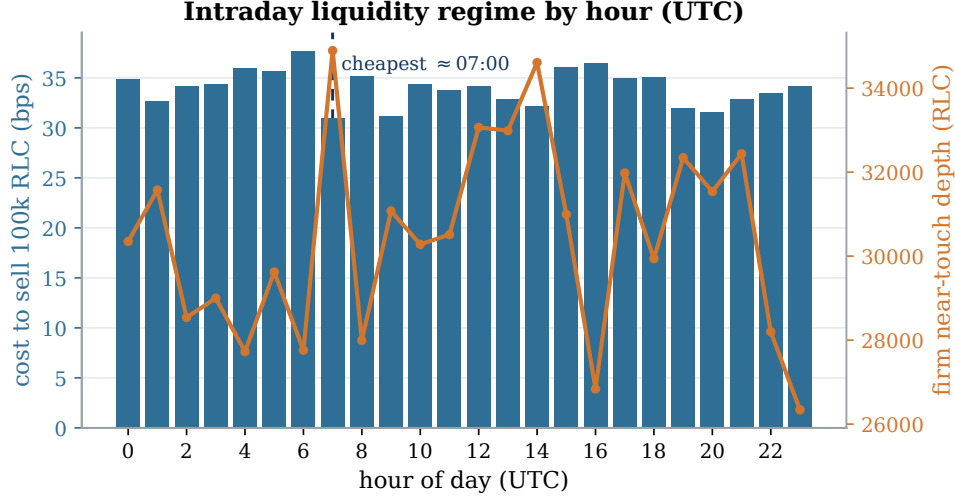


Figure 13: Intraday liquidity regime: cost to sell a fixed clip (bars) and firm near-touch depth (line) by hour (UTC). Cheapest/deepest around 07:00.

5 From Microstructure to Liquidation

We now apply the microstructure to the problem: sell RLC spot to net $\approx \$1,000,000$ USDC, *entirely on-venue* (over-the-counter block trading is unavailable), distributed across all exchanges.

5.1 Feasibility and horizon

At the reference mid, \$1M is ≈ 2.9 million RLC.¹ Three facts from Section 4 bound the program. (i) The instantaneous firm book is a median of \$315.7k (Figure 4), so no single moment clears even a third of the target; the sale must be time-distributed. (ii) Real daily volume is ≈ 830 k RLC gross (de-duplicated; ≈ 420 k one-sided buy volume), so 2.9M RLC is $\approx 3.5\times$ a day’s turnover. (iii) At a non-destructive participation of 25% of gross volume, the daily tranche is capped at ≈ 207 k RLC/day, which sets a horizon floor:

$$T_{\min} = \frac{Q}{\rho V} = \frac{2.78\text{M}}{0.25 \times 830\text{k}} \approx 13.4 \text{ days.} \quad (10)$$

Table 5 shows the participation–horizon trade-off. Faster is infeasible: the impact-optimal Almgren–Chriss burst of Eq. 6, ignoring the cap, would clear the order in ≈ 26 minutes at ≈ 107 k RLC/min— $221\times$ the measured absorption rate of 484 RLC/min—which the order book cannot supply. The ~ 13 – 14 day horizon, and therefore the exposure to RLC’s price path over that hold, is essentially *unavoidable*; without an OTC escape valve there is no way to compress it.

5.2 The speed/risk frontier and the cost band

Evaluating $C(T)$ (Eq. 8) with the estimated inputs— $\sigma = 6.1\%$ /day, permanent fraction $f = 0.67$, impact constant $Y = 0.5$, risk aversion $\lambda = 0.3$, volume $V = 830$ k/day—traces the frontier in

¹We calibrate the cost model on a representative basis $Q = 2.78$ million RLC (notional $\approx \$961,000$ at $p_0 \approx \$0.3459$); because impact and drift are expressed in basis points, the cost band is essentially invariant to using this basis versus the exact \$1M-equivalent of ≈ 2.9 million RLC, and the horizon shifts by under a day. Netting \$1M *after* costs requires oversizing the order to ≈ 3.0 – 3.2 million RLC (Step 1 of the program below).

Table 5: Participation $\rightarrow$ horizon. Days to sell 2.78M RLC at each participation rate, against gross and one-sided buy volume.

Participation ρ	vs gross volume	vs buy-side volume
10%	33.5 days	66.2 days
20%	16.8 days	33.1 days
25%	13.4 days	26.5 days

Figure 14. Because the liquidity cap forbids $T < T_{\min}$, the feasible optimum sits at the fast end of ≈ 13 –14 days for every drift scenario; drift sets the *level* of cost, not the horizon. The all-in cost is therefore a drift-conditional band (Table 6): ≈ 591 bps ($\approx \$57k$) if the price stabilises, $\approx 1,259$ bps ($\approx \$121k$) at a mild $-1\%/day$, and $\approx 2,695$ bps ($\approx \$259k$) if the in-sample $-3.5\%/day$ decline persists. The order-book-free square-root law (Eq. 9) independently puts the impact term at ≈ 152 bps at 25% participation, consistent with the frontier’s impact component and confirming the headline is not an artifact of the order-book reconstruction.

The decomposition (Figure 15) is the key economic message. Execution frictions—temporary plus permanent impact (≈ 200 bps) and fees plus cross-venue coupling (~ 30 bps)—are a minority of the cost. The timing-risk band ($\approx \pm 13\%$ at one σ over the hold) and, above all, the exogenous drift dominate. The cost of liquidating RLC is mostly the cost of *holding* a falling, volatile asset for the two weeks the market forces upon us.

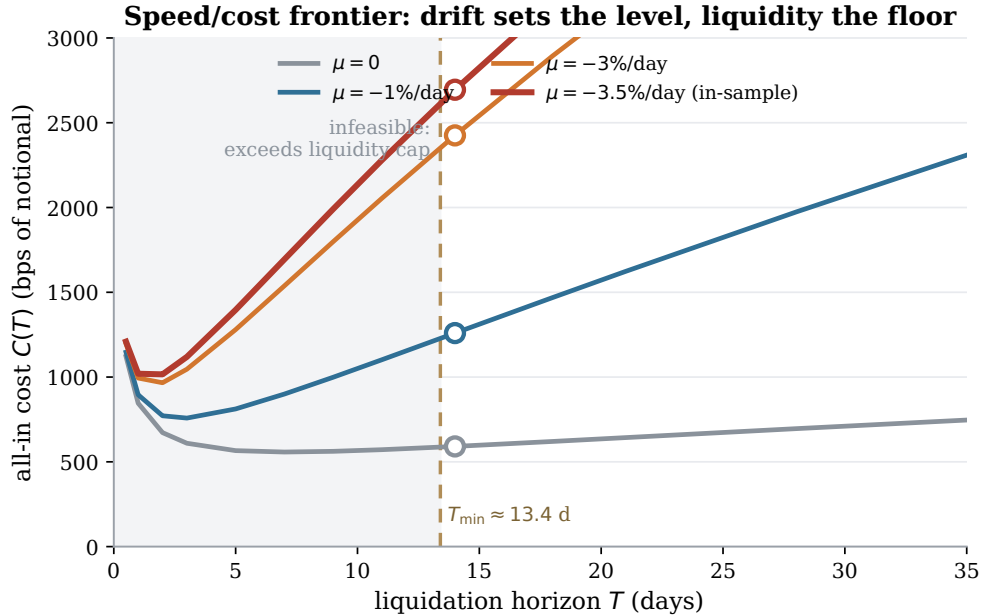


Figure 14: Speed/cost frontier $C(T)$ (Eq. 8) by drift scenario. The liquidity cap makes $T < T_{\min} \approx 13.4$ days infeasible (shaded); the feasible optimum is the fast end of ~ 14 days. Drift sets the level.

5.3 The recommended on-venue program

The microstructure dictates a specific program; Figure 16 is the schedule.

1. **Pace at the cap for ~ 14 days.** The daily participation cap (Eq. 10) binds throughout, so the

Table 6: Drift-conditional all-in cost at the feasible horizon ($T \approx 14$ d).

Price scenario	All-in cost (bps)	$\approx$ USD
Stabilises ($\mu = 0$)	591	\$57k
$-1\%/day$	1,259	\$121k
$-3\%/day$	2,425	\$233k
$-3.5\%/day$ (in-sample)	2,695	\$259k

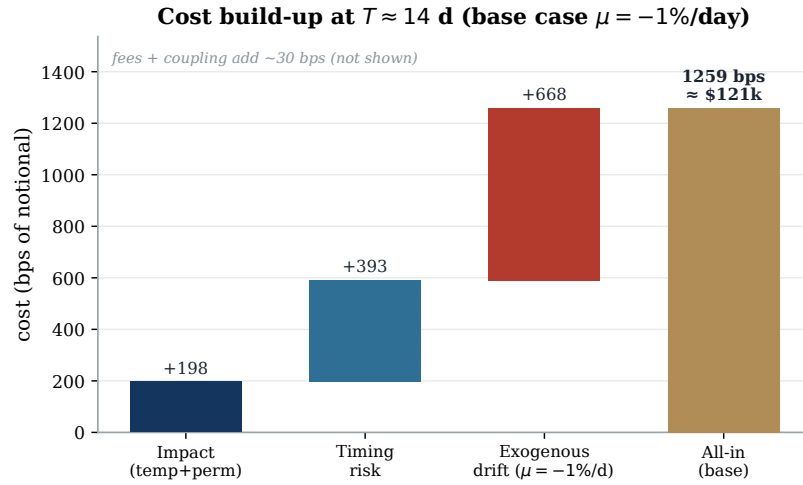


Figure 15: Cost build-up at $T \approx 14$ d (base case $\mu = -1\%/day$). Execution frictions are a minority; timing risk and exogenous drift dominate.

program runs at $\approx 207\text{k RLC/day}$. “Front-loading” operates *within* this envelope—run at the cap from day one (no ramp), and skew the intraday intensity and venue mix earlier—because the most inventory, and the most drift exposure, are carried at the start. Oversize the order to $\approx 3.0\text{--}3.2\text{M RLC}$ to net $\$1\text{M}$ after cost and the falling price.

2. **Small clips, paced under the absorption rate.** Keep clips $\lesssim 1,000$ RLC (Section 4.5) and pace them under ≈ 484 RLC/min (Section 4.4), spaced to let the cross-venue book rebuild (Figure 11). At $\sim 2,900$ clips this is necessarily an algorithm, not manual trading.
3. **Distribute across all venues, net-of-fee and MM-state-aware.** Route by firm depth, fees, and inventory state rather than concentrating on one venue: binance is deepest, firmest and its dealers are net short (natural buyers); coinbase displays size but is ghostier and its dealers are saturated long (reluctant). Split so no single venue is telegraphed, and exploit lead-lag by not always hitting the leader first.
4. **Trade the deep, cheap windows.** Weight execution toward $\sim 07:00$ and the midday–evening UTC depth (Figure 13).
5. **Treat the figures as a taker upper bound.** The costs above assume aggressive (taker) execution; a maker tilt could shave slippage at the cost of fill certainty (passive-fill probability is unobservable here).

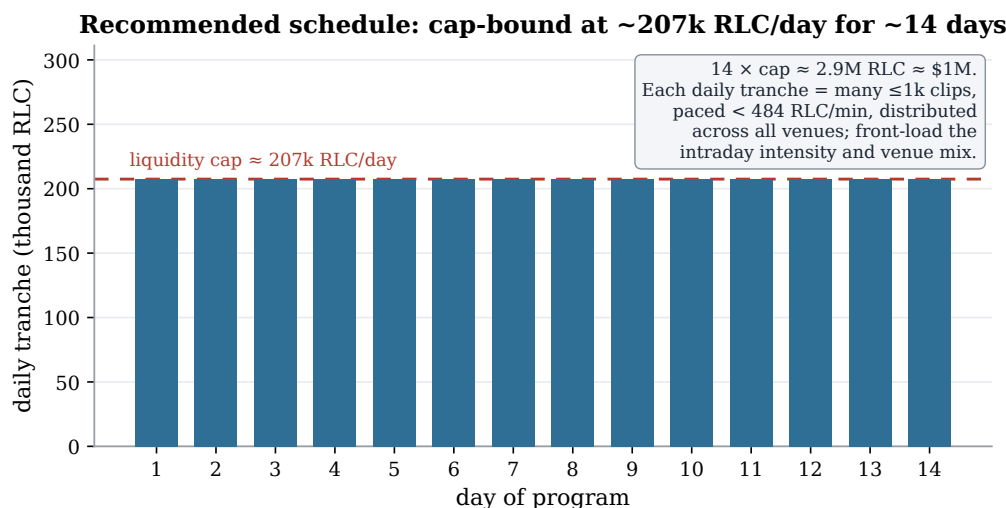


Figure 16: Recommended schedule: the daily liquidity cap binds, so the program runs at $\approx 207\text{k RLC/day}$ for ~ 14 days; each tranche is many $\leq 1\text{k}$ clips paced under the absorption rate and distributed across venues.

The single most important caveat is in Figure 17: the dominant cost is the price path, which we cannot hedge (spot-only) or escape (no OTC). Over the sample RLC fell $-3.5\%/day$ on average, but the decline decelerated—the steepest two days were -7.7% and -7.8% , the final day was $+2.1\%$ —so the realised cost will be set by which regime prevails over the actual hold, a quantity our $\sim 5\text{-day}$ sample cannot forecast.

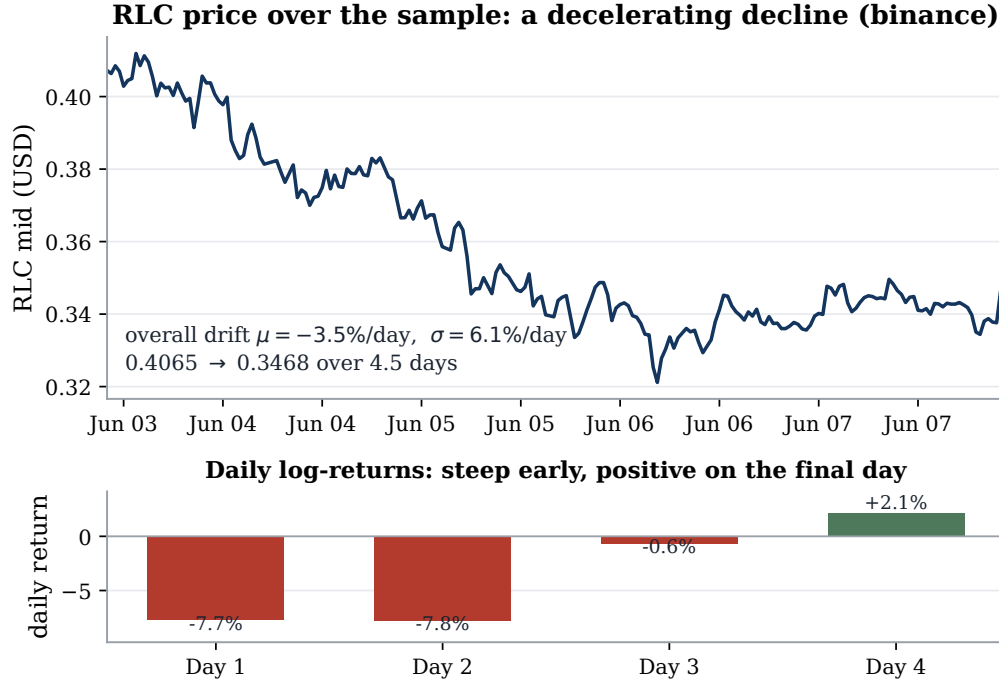


Figure 17: The dominant, un-hedgeable cost: RLC’s own price path. Overall $-3.5\%/day$ with deceleration (final day $+2.1\%$). The $-3.5\%/day$ figure is an in-sample view, not a forecast.

6 Limitations and Threats to Validity

We state limitations plainly; honesty here is part of the result. Table ?? tags every load-bearing quantity by confidence and source.

- **Short, single-regime sample.** The window is ≈ 4.5 – 4.7 days in one (declining, then flattening) regime. Microstructure estimates are stable within it but may not generalise to a stress or rally regime; intraday-timing patterns rest on few daily cycles and are directional only.
- **Drift is an in-sample view, not a forecast.** The $-3.5\%/day$ decline drives the cost band but cannot be projected forward from five days; we report cost as a scenario band, never a point.
- **Large-clip impact and contagion scaling are not identified.** The prints in the sample are small; impact and contagion at execution clip sizes are bounded (swept band, square-root cross-check), not measured.
- **MM inventory is L2 inference.** With no order IDs we infer dealer behaviour from quote and trade dynamics; the inventory *level* and the flip point are not identified (Section 4.4); the absorption rate and inventory dispersion are.
- **Spot-only; no hidden liquidity or queue position.** Perpetual venues and any hedge are out of scope; iceberg/hidden depth and queue position are invisible in public L2 data, so realised passive fills may differ.

7 Conclusion

The empirical core of this paper is a portrait of a thin altcoin’s microstructure: a fragmented, thin, sharply convex consolidated book whose displayed depth is inflated by heavy quote churn; a concentrated, adverse-selection-aware dealer population whose inventory is venue-split (binance short, coinbase long) with a well-identified absorption rate (≈ 484 RLC/min) but an unidentified level; and tightly coupled venues in which binance leads and the others follow within half a second, with cross-venue contagion that becomes material above $\sim 1,000$ RLC and recovers over tens of seconds. The liquidation follows as an application. Because the firm book absorbs only $\approx \$0.32$ M at any instant and real volume is ≈ 830 k RLC/day, selling \$1M is a ≈ 13 –14 day, multi-venue, small-clip program; with no OTC route the horizon and its drift exposure are unavoidable. The least-cost feasible program runs at the participation cap, front-loaded within it, in $\lesssim 1$ k-RLC clips paced under the absorption rate, distributed across venues by depth, fees and dealer inventory state, and timed into the deep windows. Its all-in cost is a drift-conditional band of ≈ 591 – $2,695$ bps, of which execution frictions are only ~ 230 bps; the dominant, un-hedgeable term is RLC’s own price path over the hold. The actionable conclusion is therefore as much about *risk* as about *execution*: optimise the on-venue mechanics, but recognise that the outcome is governed by where RLC trades over the coming fortnight, which the data can bound but not predict.

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